# Yash Chauhan

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*Final Year Int. MSc. Physics Major*

## Experience

**Weizmann Institute of Science - Yuval Ronen Lab (Quantum Materials Transport),** *June 2026 - Aug 2026*
*Kupcinet-Getz Summer Intern* (Prof. Yuval Ronen) | Rehovot, Israel
- Worked on understanding **Quantum Hall Interferometry** and Anyon Braiding.
- Prepared stacks of bilayer graphene.

**Liaoning Academy of Materials - Institute of Quantum Materials and Devices,** *Visiting Student* *May 2026 - June 2026*
(Prof. Zheng Vitto Han) | Shenyang, China
- Learned how to **fabricate 2D materials** and devices.
- Prepared a stack of twist $MoS_2$.
- Also explored **automation** in 2D Materials research.

**National Institute for Materials Science - Surface Quantum Phase Materials Group,** *Winter* *Dec 2025 - Jan 2026*
*Intern* (Prof. Takashi Uchihashi) | Tsukuba, Japan
- Invited to return to continue my summer internship work; collected additional data that resulted in a **publication in PRB**.

**RPTU Kaiserslautern - Theoretical Quantum Optics Group,** *Summer Intern* (Prof. Michael *Jul 2025 - Feb 2026*
Fleischhauer) | Kaiserslautern, Germany
- Worked on deriving **Effective Hamiltonains** for Soliton solutions in a 2D Topological Insulator with interactions.

**National Institute for Materials Science - Surface Quantum Phase Materials Group,** *Summer* *May 2025 - June 2025*
*Intern* (Prof. Takashi Uchihashi) | Tsukuba, Japan
- Grew **atomic-layer** Indium **superconductors** on Si substrates via sputtering; studied **2D superconductivity**.
- Performed transport measurements using a **UHV STM + Transport** setup (0-8T, 300mK).
- Investigated **Josephson vortex** formation and dynamics under perpendicular and parallel magnetic fields.

**NISER - Laboratory of Novel Magnetism and Spintronics,** *Research Volunteer* (Prof. Ajaya *Aug 2024 - Present*
Kumar Nayak) | Bhubaneswar, India
- Currently working on my **Master's Thesis** in collaboration with Prof. Karthik V. Raman's lab.
- Initially involved in synthesizing potential **altermagnetic** rare earth compounds via arc-melting and self-flux methods (now paused); currently learning thin film deposition techniques like PLD and sputtering.

**Raman Research Group, Tata Institute of Fundamental Research,** *Summer Intern* (Prof. Karthik *June 2024 - Aug 2024*
V. Raman) | Hyderabad, India
- Worked in Fabricating Thin Epitaxial Films of $(\mathbf{Bi_xSb_{1-x}})_\mathbf{2}\mathbf{Te_3}$ topological insulator.
- Acquired hands-on experience in **MBE, XRD, AFM, ICE** cryostat, transport measurements.
- Also explored theoretical condensed matter and many body physics - Topological matter, **Neutral Atom Quantum Simulation** using Rydberg Arrays and Altermagnetism.
- Went back during the winter break (*Dec, 2024*) to acquire hands-on experience in operating **STM**. Studied the surface of $KV_3Sb_5$, a kagome superconductor.

**Quantum Denoising for Brain-Computer Interfaces/Quantum SVM,** *Intern* (Dr. Grégoire Cattan) *Feb 2024 - Feb 2025*
| Remote
- Investigated self-supervised learning for **EEG**-based passive Brain-Computer Interfaces (pBCIs)
- Explored **Quantum Autoencoders** for signal denoising.
- Abstract accepted at the **$5^{th}$ International Neuroergonomics Conference**
- Compared Quantum and RBF kernels for the Diagnosis of **Parkinson** using **QSVM**

**NISER - Color Spanning Sets,** *Research Volunteer* (Dr. Abhishek Sahoo) | Bhubaneshwar, India *Dec 2023 - May 2024*
- Studied the fixed parameter tractability of some of the **NP-hard** Color Spanning Set problems.

**AICH Lab - MDNA,** *Research Volunteer* (Prof. Palok Aich) | Bhubaneshwar, India *June 2023 - Oct 2023*
- Worked on understanding the **HOMO-LUMO** band gap of **metal-doped DNA** structures.
- Performed ab-initio **DFT** calculations using ORCA and modelled the structures on Avogadro.

**Indian Statistical Institute,** *Student Intern* (Dr. Ramij Rahaman) | Kolkata, India *Mar 2023*
- Learnt about several **QKD** protocols - implementations and attacks.
- Worked on understanding the security proof of the **BB84** protocol.

**National Physical Laboratory,** *Winter Intern* (Dr. H.K. Singh) | New Delhi, India *Dec 2022*
- Learnt about **Thin Film** depositions using the **RF Magnetron Sputtering** System.
- Participated in the fabrication of **PCMO** on substrates of Lanthanum Aluminate and Strontium Titanate

## Publications

### Preprints

- Kar, R., **Chauhan, Y.**, Sahu, A. (2026) Single-Exponential Fixed-Parameter-Tractable Algorithms for Euclidean Color-Spanning Set Problem (manuscript under preparation)

**Published**

- Qian, W., **Chauhan, Y.**, Nemoto, R., Sagisaka, K., Yoshizawa, S., & Uchihashi, T. (2026). Anisotropic transport of Josephson vortices in atomic-layer superconductors on vicinal surfaces. Phys. Rev. B, 114, 055422.
- Cattan, G., Olmos, J., **Chauhan, Y.** et al. Parkinson disease classification: a comparison of quantum and RBF kernels using support vector machine. Quantum Mach. Intell. 7, 64 (2025). https://doi.org/10.1007/s42484-025-00286-0
- Ahmed Mostafa, **Yash Chauhan**, Wasif Ahmed, Grégoire Cattan, Marius Klug. Quantum Denoising for BCI: A Study. 5th International Neuroergonomics Conference, University of Bordeaux, Jul 2024, Bordeaux, France. pp.494-499. (hal-05020004)

## Education

8.59/10 **Integrated MSc. in Physics**, *National Institute of Science Education and Research* | Bhubaneshwar, India *2022-27 (Present)*

92.30 % **Std. XII, CBSE**, *Career Point Gurukul* | Kota, India *2022*

94.50 % **Std. X, ICSE**, *Sharda Mandir School* | Goa, India *2020*

**Achievements:** DAE DISHA Scholarship | WOMANIUM Quantum Scholarship | Tensorflow Education Stipend | Travel Award - $5^{th}$ International Neuroergonomics Conference
**Conference:** National Quantum Science and Technology Symposium (QETCI)
**Research Interest:** Topological Quantum Matter, Quantum Hall Physics, 2D Materials, Circuit QED, Spintronics, Altermagnetism, Many-Body Physics, Quantum Algorithms, Quantum Information, Systems Biology, Mathematical Modelling, Quantum Information Science

## Skills

**Experimental** MBE, STM, AFM, XRD, ICE Cryostat, Hall Bar Measurements, Transport Measurements, Wire Bonding/Scratching, Arc-Melting, Self-Flux Method, Sputtering, Mechanical Exfoliation, Dry Transfer Stacking

**Programming** Python, C/C++, Javascript, Git, Bash, LaTeX, HTML, Julia, CSS, Java

**Software** Linux, Github, ORCA, Avogadro, LabVIEW, Gwyddion, ORIGIN | qiskit, cirq, tensorFlow, bloqade.jl, yao.jl, Data Analysis

**Certifications**
- IBM Certified Associate Developer - Quantum Computation using Qiskit v0.2X
- DeepLearning.AI Tensorflow Developer Professional Certificate
- Qiskit Global Summer School 2023 | Qbronze, QNickel, QSilver, QMercurcy - QWorld
- TCS Quantum Challenge 2023

## Projects

**The Parton Route to Quantum Spin Liquids** *May 2026*

NISER, P451 Advanced Solid State Physics: Prof. Anamitra Mukherjee

- Learned about **Parton Mean Field Theory** and **Slave Fermion** representation to solve the Heisenberg Model, Kitaev Model and also to analyze the Kitaev-Heisenberg Model.

**Topolectric Circuits** *Present*

NISER

- Exploring and implementing **non-Hermitian** and **topological physics** in electrical-circuit platforms (topolectrical circuits), including circuit realizations of lattice models and frequency-domain characterization of edge/localized modes.

**Hidden Polarization and Obstructed Topological Superconductivity in a Spinful SSH Analogue** *Nov 2025*

NISER, P468 Magnetism & Superconductivity: Prof. Colin Benjamin

- Re-derived and numerically reproduced the main results of Jin *et al.* (Phys. Rev. B 109, L241101 (2024)) for a 2D spinful SSH analogue hosting hidden polarization and obstructed **topological superconductivity**.
- Implemented tight-binding and BdG calculations to verify bulk spectra, Wannier/Wilson-loop polarization signatures, and midgap edge modes in ribbon geometries.
- Modeled three-terminal normal–superconductor–normal transport and reproduced asymmetric nonlocal conductance features.

**Ising Games on Graphs Revisited** *Nov 2025*

NISER, P468 Magnetism & Superconductivity: Prof. Colin Benjamin

- Reproduced and extended the core analytical framework of Leonidov et al. *"Ising games on graphs"*.
- Recovered equilibrium magnetisation curves and metastability / hysteresis (spinodals) in mean-field: obtained the Curie–Weiss relation on complete graphs and quantified how degree heterogeneity reshapes the hysteresis window under an annealed configuration-model approximation.
- Derived macroscopic Fokker–Planck / drift–diffusion dynamics and validated with simulations under driving protocols (step-quench and slow/fast field sweeps).

**Understanding Algorithmic Learning Through Transformer Shortcuts** *April 2025*

NISER, CS461 Advanced Machine Learning: Prof. Anup Kumar Bhattacharya

- Based on the paper *"Transformers Learn Shortcuts to Automata"* by Bingbin Liu et al., conducted a detailed study of **how shallow Transformers simulate automata-like logic**.
- Explored shortcut solutions that enable Transformers to bypass sequential processing using parallel architectures.
- Investigated theoretical tools like Krohn–Rhodes decomposition and circuit complexity, and evaluated empirical robustness under varying training conditions.

**A Quantum Approach to Option Pricing** *Nov 2024*

NISER, P346 Computational Physics: Prof. Colin Benjamin

- Replicated key results from the seminal paper "Option Pricing Using Quantum Computers" by Stamatopoulos et al., showcasing the application of **Quantum Amplitude Estimation** for option pricing.
- Designed and implemented quantum circuits to price vanilla, basket, and path-dependent options using Qiskit, analyzing potential quadratic speed-up over classical Monte Carlo methods.
- Investigated challenges in applying quantum algorithms to real-world financial problems, emphasizing hardware requirements and algorithm scalability for quantum finance applications.

**Optimising Fleet Allocation** *Nov 2023 - Mar 2024*

TCS Quantum Challenge 2023, Remote

- Conducted background research on the **Airline Scheduling** problem and performed preliminary experimental validation as part of Phase I.
- Worked with a team of four members accompanied by technical and academic advisors on **Quantum Optimization** using **Amazon Braket**.
- Reached the Final stage and delivered a presentation to a panel consisting of eminent academicians from Oxford and CEOs of several airline companies.

## WORKSHOPS/EVENTS

**13th Annual International Hackathon for Social Good** *Apr 2025*

New York University, Abu Dhabi (NYAUD)

- Developed a proof-of-concept pipeline for cancer risk minimization using radiotherapy, integrating beam angle optimization in hardware setups.
- Designed a frontend system for cancer detection from radiological data as part of Team QurAI.

**QUIKSTART x QML** *Feb 2025*

RPTU Kaiserslautern, Germany

- Selected to attend a winter school on **Quantum Machine Learning** and received a grant of **€1590** for participation.
- Presented a poster titled *Quantum Denoising for BCI* during the school.

## Extra-curricular

**Chess Club** Organised intra-college events, friendly matches and promoting chess
**Tennis Club** Represented the college in tournaments
**Physics Club** Started a subgroup focused on mathematical physics

## Additional Coursework

2023 **Quantum Algorithms and Cryptography**, IIT Madras, NPTEL
2023 **Graph Theory**, IISER Pune, NPTEL
2023 **Cryptography I**, Stanford Online, Coursera
2024 **Metric Spaces**, NISER, In Person
2025 **Introduction to Manifolds**, NISER, In Person

## Languages

**English** Professional proficiency
**Hindi** Native proficiency
**German** Currently Learning (Beginner)